

Online Experiments for Language Scientists

Lecture 9: Zipf's Law of Abbreviation

Kenny Smith

kenny.smith@ed.ac.uk

Assessment 2 Q&A

- Due on 5th December
- Read the assignment brief (linked from <https://kennysmithed.github.io/oels2024/>)
- We can help with basic coding stuff in extra drop-in labs (see times on github course page)
- **Happy to answer questions now, or email me (be aware: I am travelling 25th to 30th November, responses may be delayed)**
- No questions after 10am on Monday 2nd December (other than in drop-ins)

Kanwal et al (2017)

Kanwal, J., Smith, K., Culbertson, J., & Kirby, S. (2017). Zipf's Law of Abbreviation and the Principle of Least Effort: Language users optimise a miniature lexicon for efficient communication. *Cognition*, 165, 45-52.

A dyadic interaction experiment using a miniature language

- Does Zipf's Law of Abbreviation arise from competing pressures to communicate accurately but efficiently?



Jasmeen Kanwal
(now at St Andrews)



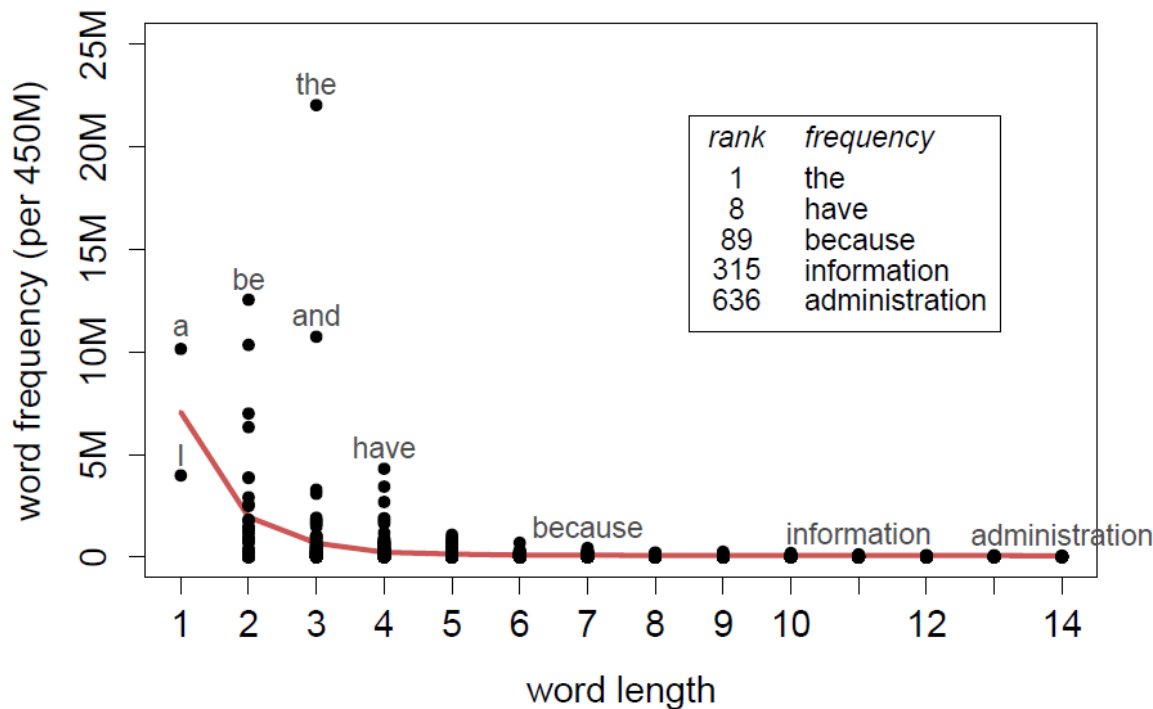
Jenny Culbertson
(Edinburgh)

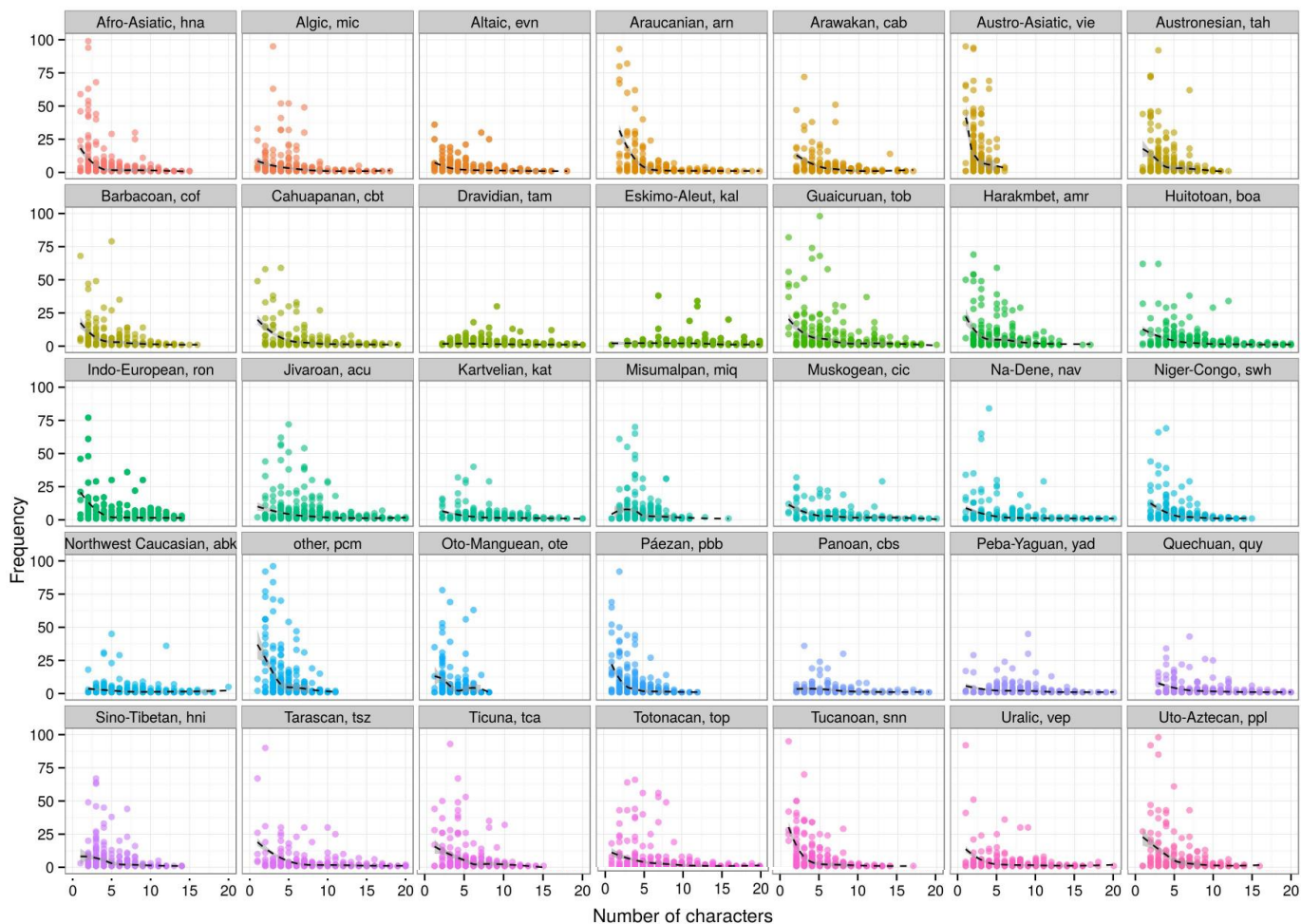


Simon Kirby
(Edinburgh)



Zipf's Law of Abbreviation: frequent words are short





Manipulating communicative need and production effort



zop x 4
zopekil x 4



zop x 12
zopoudon x 12

- Communicative task **or** asocial recall task
- Production effort depends on length **or** it doesn't

Waiting for partner

Choose a name to describe this object to your partner.

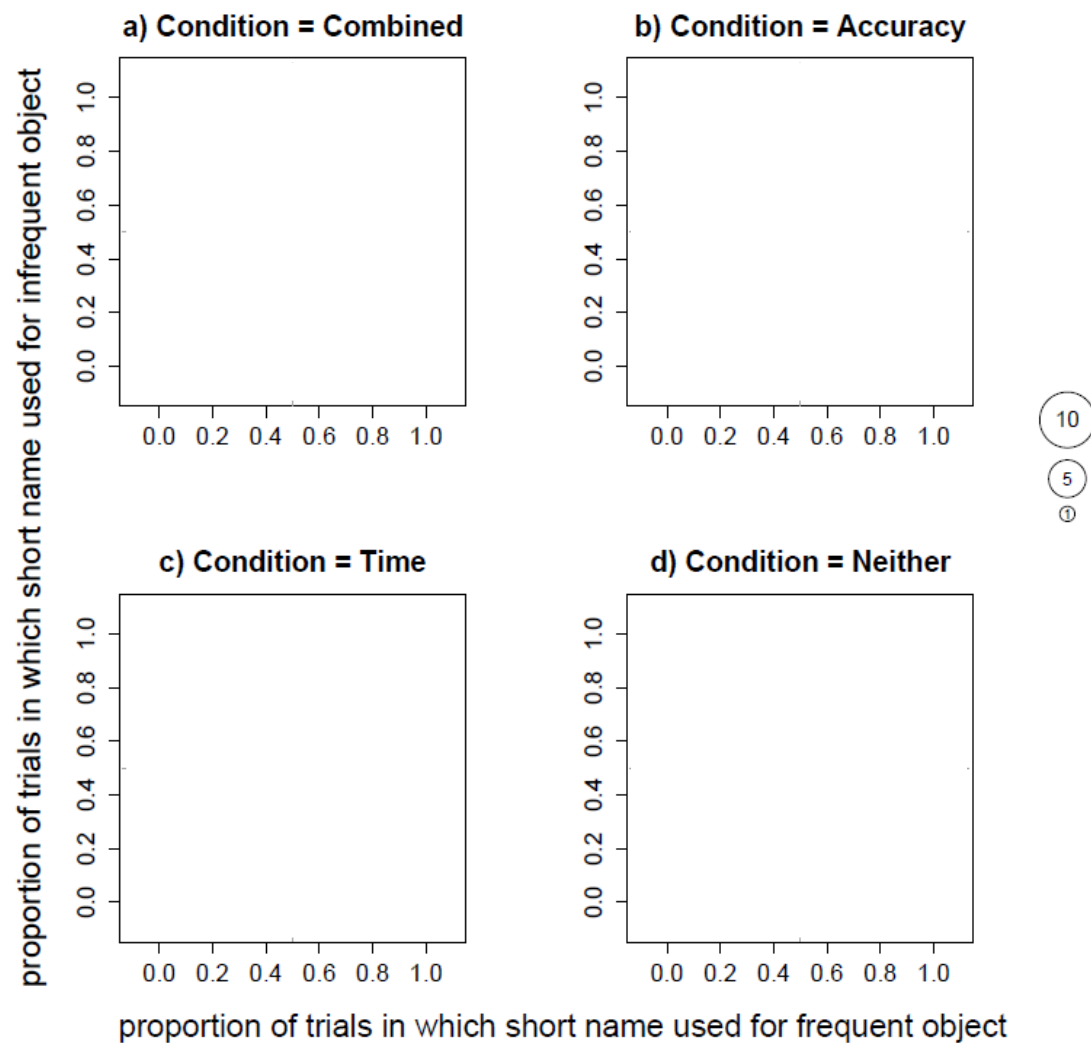


zopekil

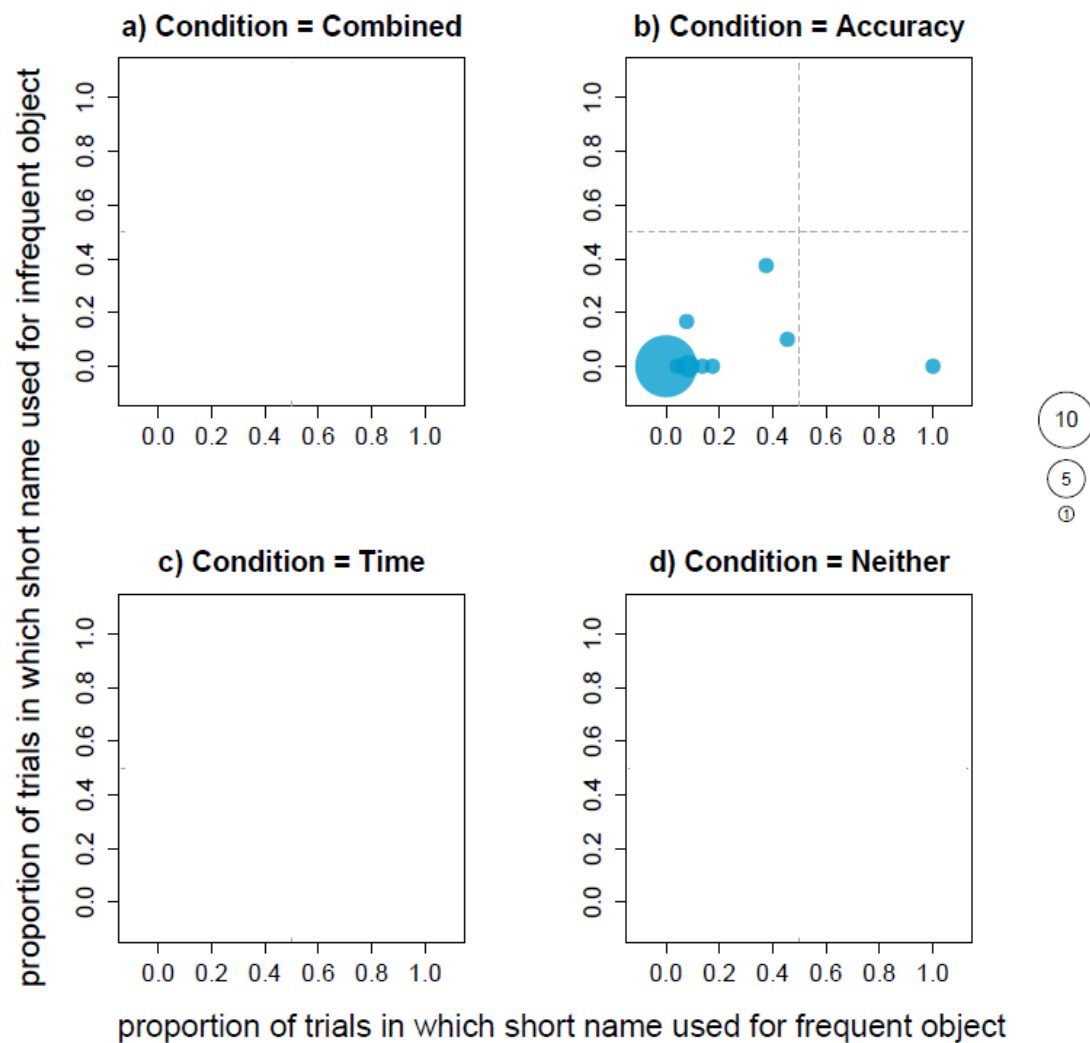
zop

		Production effort?	
		yes	no
Communication pressure	yes	Combined condition	Accuracy condition
	no	Time condition	Neither condition

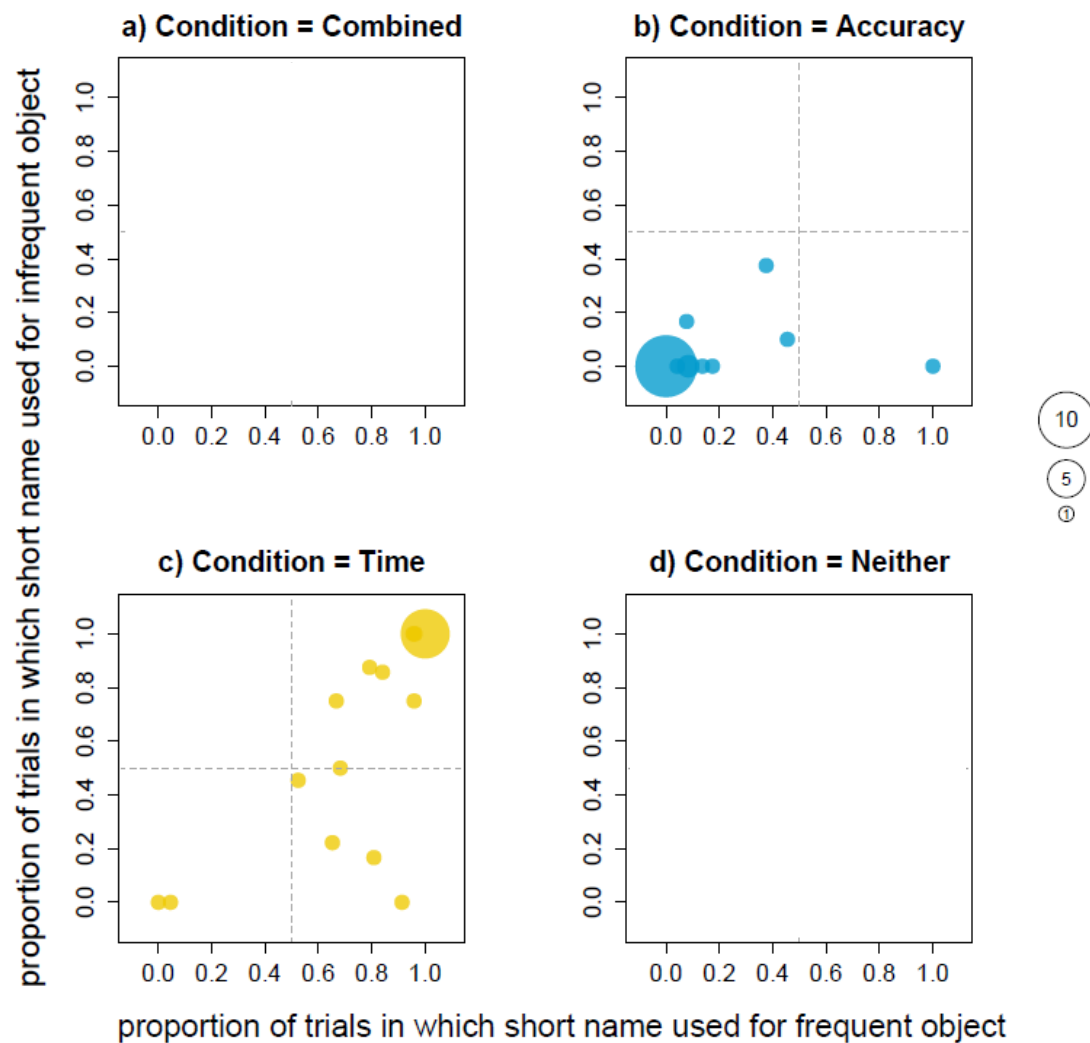
Results



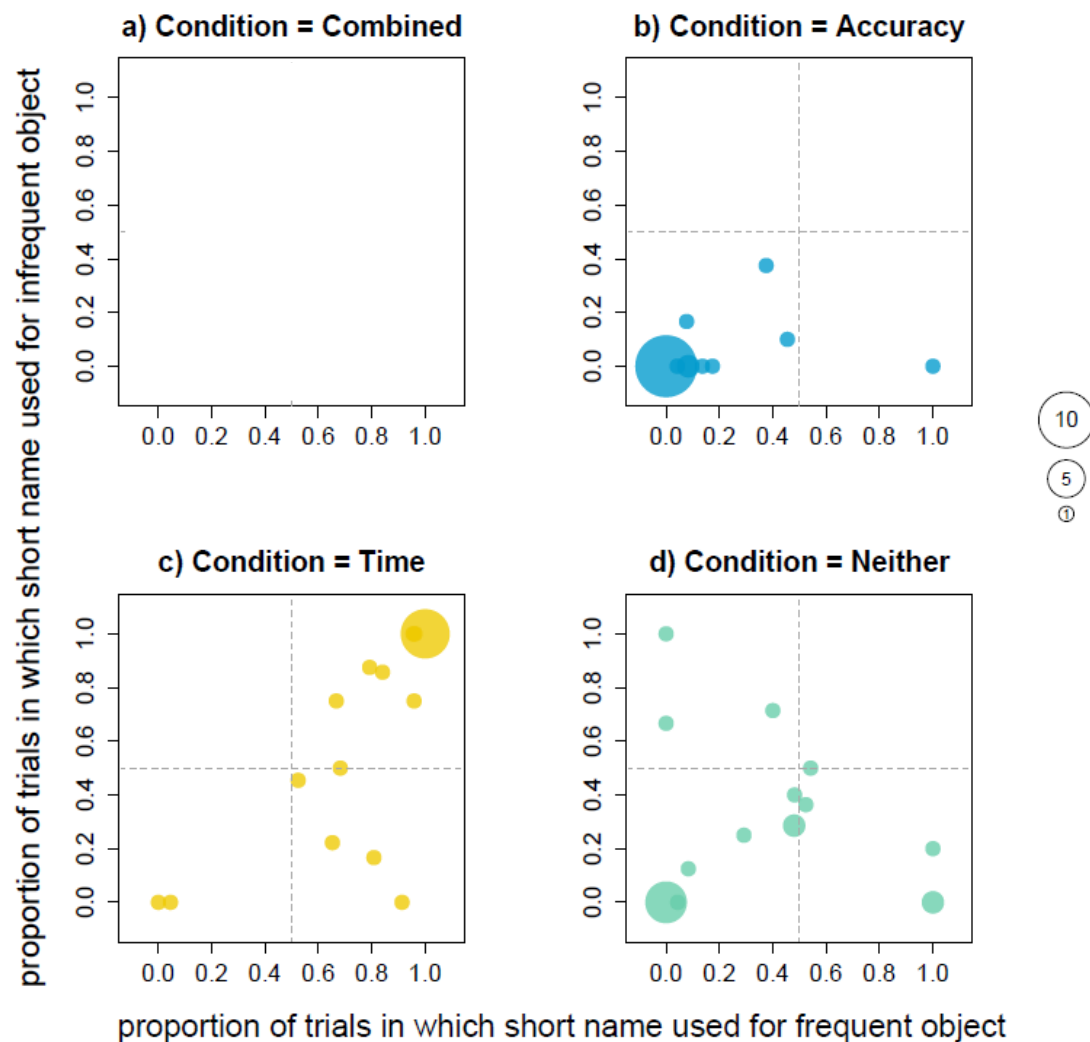
Results



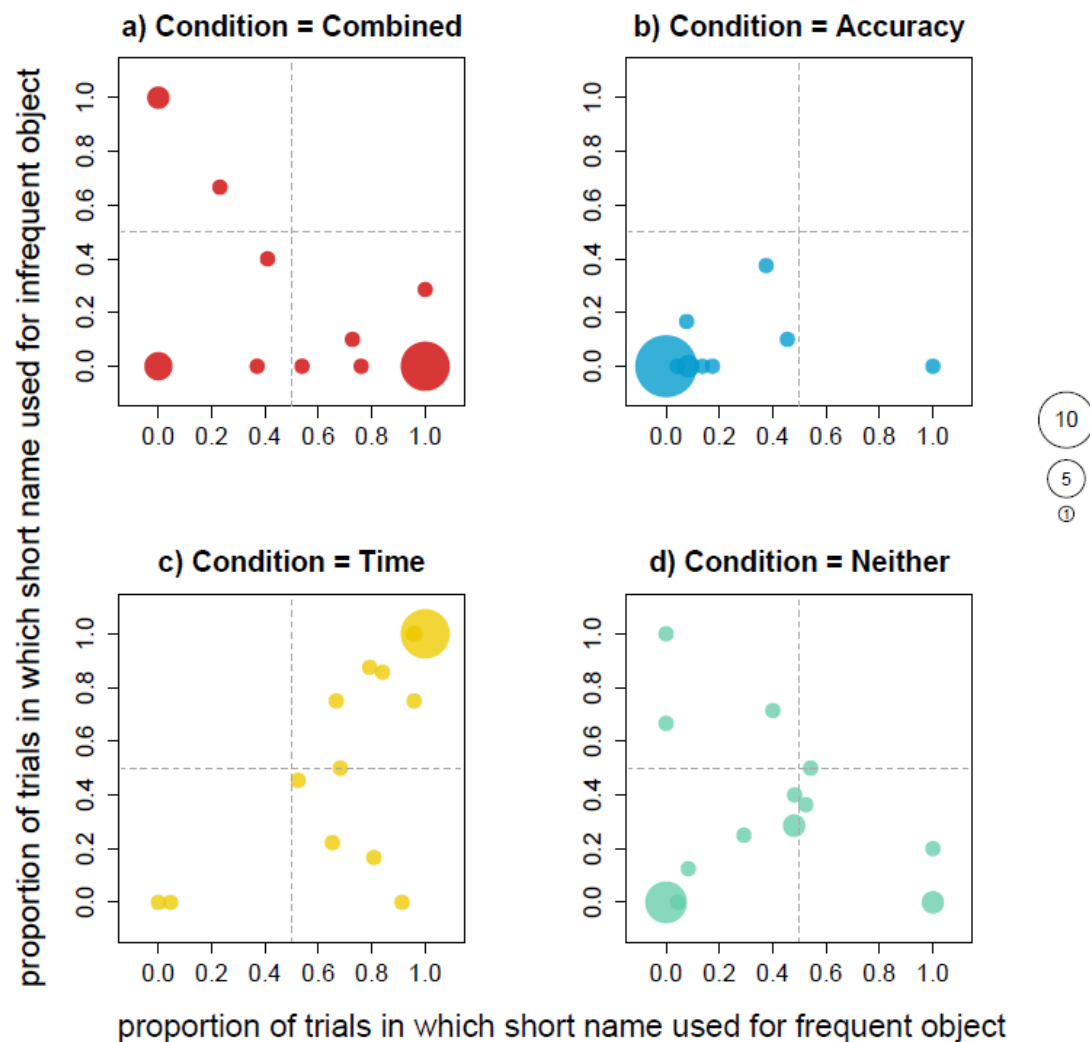
Results



Results



Results



Kanwal et al.'s conclusions

Zipf's law of abbreviation is a result of jointly optimizing competing pressures to communicate both accurately and efficiently

- Just accuracy or just efficiency is not enough

Time for additional Q&A/discussion on this week's reading

Next up

Wednesday, 9am: our last lab!

- A dyadic interaction experiment

Subsequently: optional drop-ins for debugging help with final assessment code, see [github page](#) for details